Strategy Engine for a UAV Fleet using a Genetic Algorithm

Artificial Intelligence – BSCS 5, Bahria University Lahore Campus – Assignment 2 (CLO4)

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1. Overview

Three UAVs survey a 50×50 grid with disaster hotspots, hazard pockets, no-fly zones, and hidden targets. The engine decides each drone's deployment, waypoints, speed, and sensor schedule. Objectives conflict: full-power sweeps maximise coverage but drain battery and cross hazards; staying safe and cheap leaves hotspots unseen. There is no single optimum, only trade-offs. A Genetic Algorithm fits because the landscape is non-convex, partly discrete, and only evaluable by simulation.

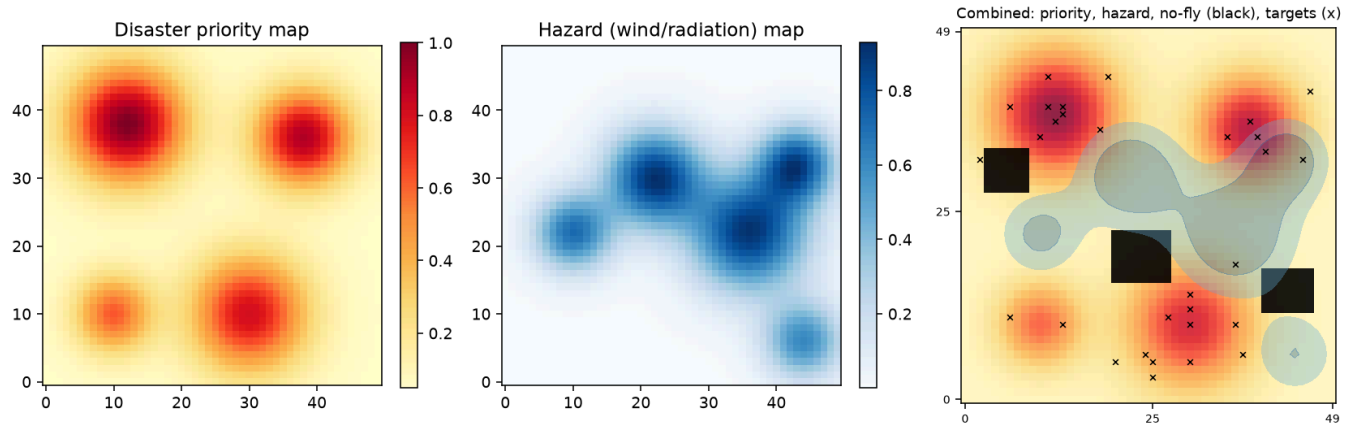


Figure 1: Priority map, hazard map, and combined view with no-fly zones (black) and hidden targets (×).

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2. Chromosome Design

Hybrid real/binary multi-segment chromosome. For each drone d and segment k = 1..8:

Table 1: Chromosome gene blocks

Gene block	Type	Meaning
deploy[d] (x,y)	real	deployment cell
wp[d,k] (x,y)	real	waypoint k
speed[d,k]	real in [0.5, 2]	cruise speed on segment k
power[d,k]	real in [0, 1]	sensor power (radius 2–6 cells)
active[d,k]	binary	sensor ON/OFF on segment k

Total = 3 × (2 + 5·8) = 126 genes.

Rationale: real coordinates let spatial operators act smoothly; sparse waypoints with interpolated segments keep the genome short but describe long routes; speed/power/schedule sit in the same segment as the

waypoint so one crossover preserves both the route piece and how it is flown; the binary schedule lets the GA turn the sensor off in transit to save energy.

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3. Fitness

Each segment is sampled at 16 points; sensor discs are stamped on the grid; energy, time, hazard, and violations are accumulated.

- Coverage = priority-weighted fraction of covered cells.
- Detection = mean over targets of q = min(1, (0.4+0.6·power)·1.2/speed) if inside an active footprint, else 0.
- Energy = Σ[length·(0.5 + 0.5v^2) + 2·$power^2$·time·ON] / total battery.
- Time = slowest drone's flight time / 250.
- Hazard = Σ hazard(cell)·time-in-cell / 100.

F = 0.35·Cov + 0.25·Det − 0.15·E − 0.10·T − 0.15·H − λ(g)·V, λ(g) = 2·(1 + 3g/G).

Weights: benefits sum to 0.60, costs to 0.40 — mission success dominates but costs matter. Coverage outranks detection because it is broader and partly implies detection. Energy and hazard (0.15 each) are the real operational risks; time is lowest (0.10) because the range limit rarely binds. All terms are normalised to roughly [0, 1].

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4. Genetic Operators

- Initialisation: 50 % random-walk routes, 50 % uniform-random point sets.
- Selection: tournament of size k, with 2 elites.
- Crossover (rate 0.85), one of three chosen at random: (i) route swap; (ii) segment swap of a waypoint sub-sequence with speed/power/schedule; (iii) geometric convex blend of coordinates and continuous genes, uniform crossover for bits.
- Mutation: Gaussian waypoint/deployment shift (σ annealed 6 → 0.8 cells); jump mutation re-sampling uniformly or from the priority map; waypoint swap and sub-route reversal; speed/power perturbation; sensor-bit flip.
- Premature-convergence control: after 10 stagnant generations, mutation doubles and 10 % random immigrants replace the worst.
- Termination: 150 generations, or 60 stagnant generations.

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5. Constraints

Hard: path ≤ 150 cells/drone, battery ≤ 200/drone, no-fly zones forbidden.

- Repair: clip bounds; move waypoints out of no-fly cells; contract over-long routes by 0.98·R/length.
- Dynamic penalty: V = Σmax(0, len−150)/150 + Σmax(0, E−200)/200 + 10·(length inside no-fly)/150, multiplied by λ(g), which grows 4× over the run.
- All 5 GA runs and 5 random-search runs returned feasible solutions.

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6. Results

Best fitness rises from 0.05 to 0.38; diversity falls 0.28 → ~0.10 without collapsing. Early routes cross hazards; later generations deploy drones beside hotspots, shorten transit legs, switch the sensor off in transit, and route around hazards. Final: coverage 0.73, detection 1.00, energy 0.48, time 0.32, hazard 0.13, no violations.

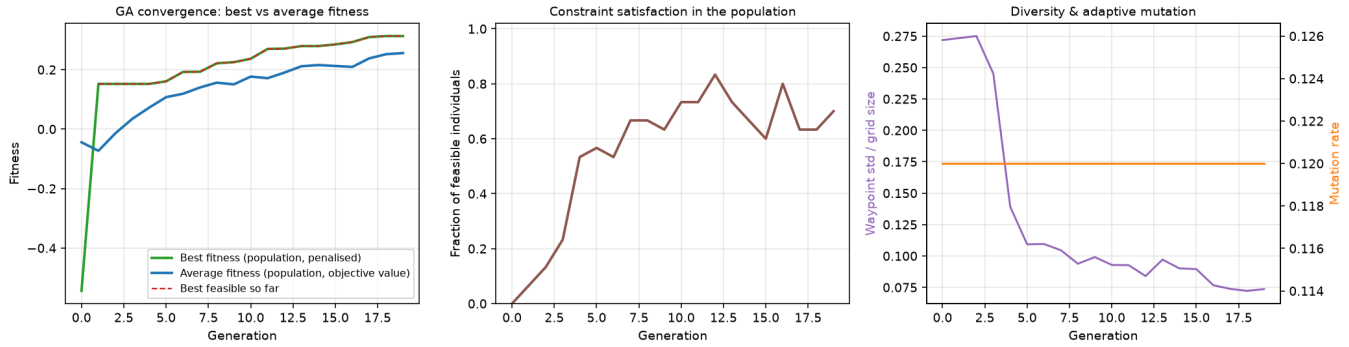


Figure 2: GA convergence — best vs average fitness, constraint satisfaction, and diversity.

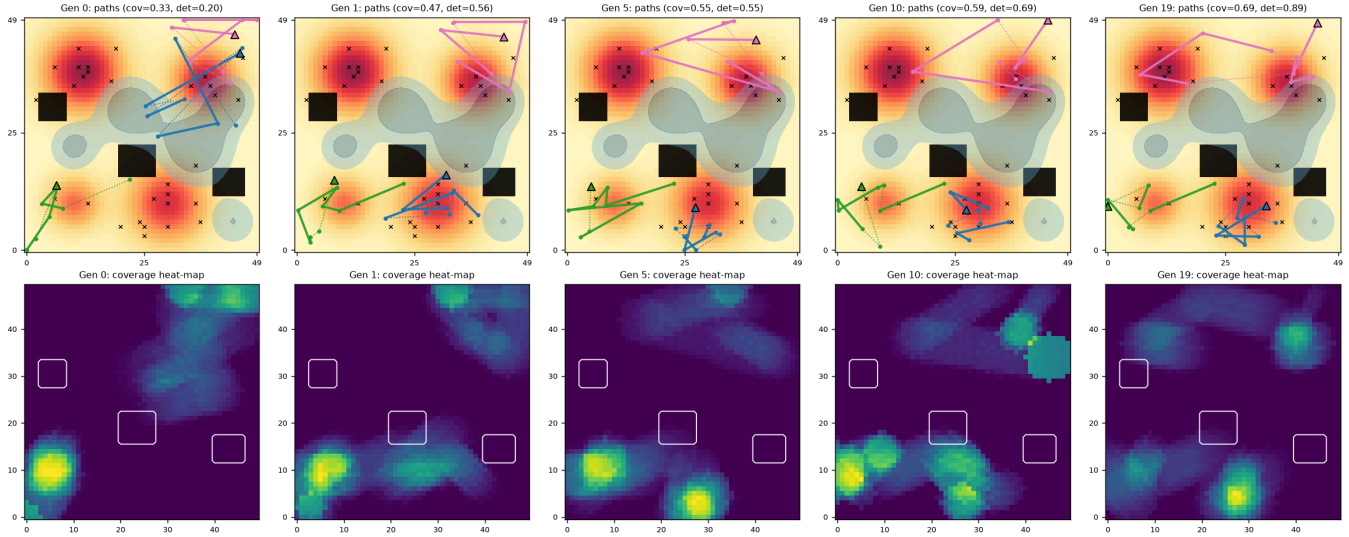


Figure 3: Best paths and coverage heatmaps at generations 0, 12, 37, 75, 149.

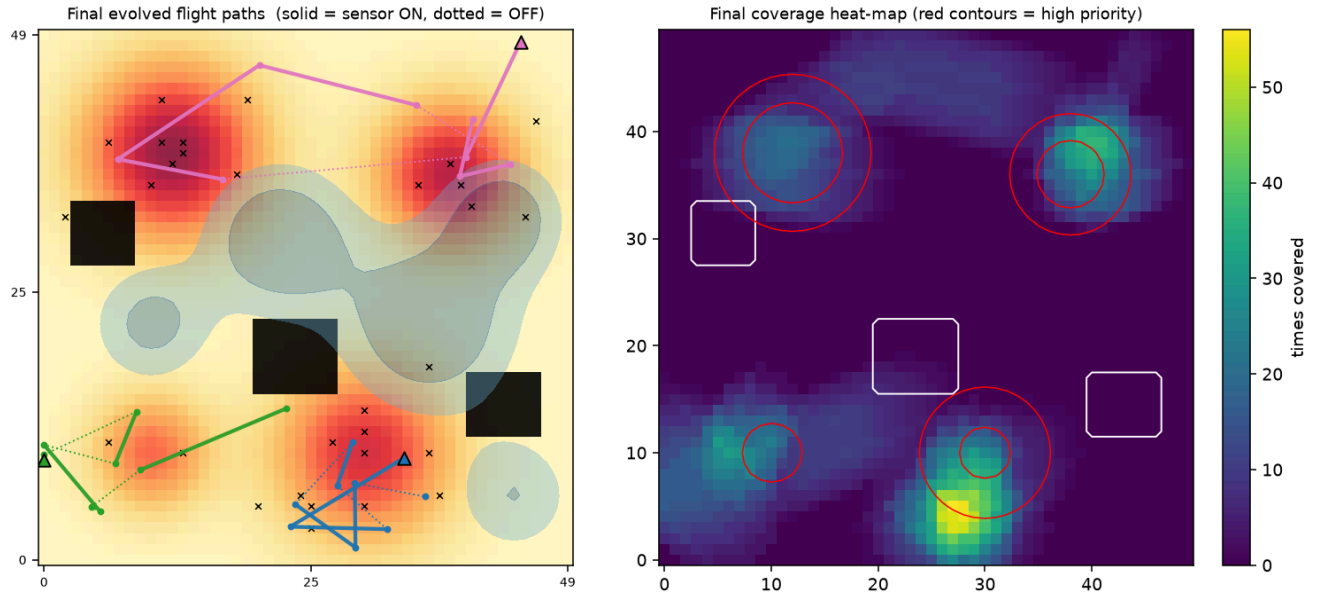


Figure 4: Final evolved paths (solid = sensor ON, dotted = OFF) and coverage heatmap.

7. Hyperparameter Sensitivity

Table 2: Sensitivity (3 seeds, 80 generations)

Parameter	Value	Fitness (mean ± std)	Diversity	Gens to 95%
Population	20	0.312 ± 0.026	0.083	47
	40	0.331 ± 0.007	0.107	49
	80	0.369 ± 0.003	0.105	50
Mutation	0.02	0.420 ± 0.010	0.020	50
	0.12	0.369 ± 0.003	0.105	50
	0.30	0.235 ± 0.015	0.172	32
Tournament k	2	0.273 ± 0.038	0.116	55
	4	0.369 ± 0.003	0.105	50
	8	0.372 ± 0.011	0.105	29

Findings: larger populations give better and more stable results; mutation rate has the largest effect (0.02 best for fitness, 0.30 noise-dominated) — recommend 0.05 for balance; k = 2 too weak, k = 8 converges fastest but loses diversity.

Table 3: Weight-profile trade-offs

Profile (wC, wD, wE, wT, wH)	Cov	Det	Energy	Time	Hazard
Balanced (.35, .25, .15, .10, .15)	0.73	1.00	0.48	0.32	0.13
Coverage-first (.50, .30, .05, .05, .10)	0.83	0.99	0.79	0.36	0.41
Energy-saver (.25, .15, .35, .15, .10)	0.59	0.83	0.32	0.23	0.20
Safety-first (.25, .15, .10, .10, .40)	0.65	0.87	0.48	0.22	0.03

Coverage-first gains 10 points of coverage but spends 65 % more energy and ~3× the hazard exposure; Safety-first nearly eliminates hazard at the cost of 8 coverage points. The engine returns a solution consistent with the operator's priorities, not a universal optimum.

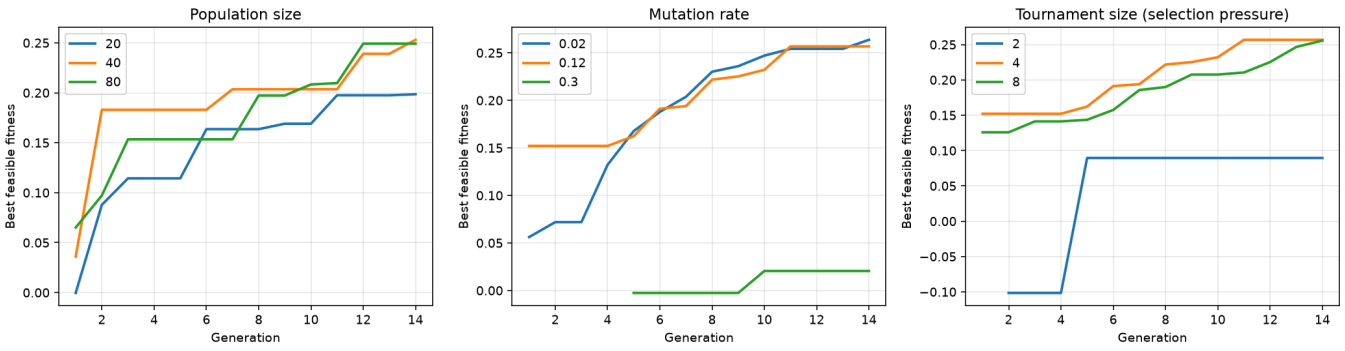


Figure 5: Hyperparameter sensitivity (population size, mutation rate, tournament size).

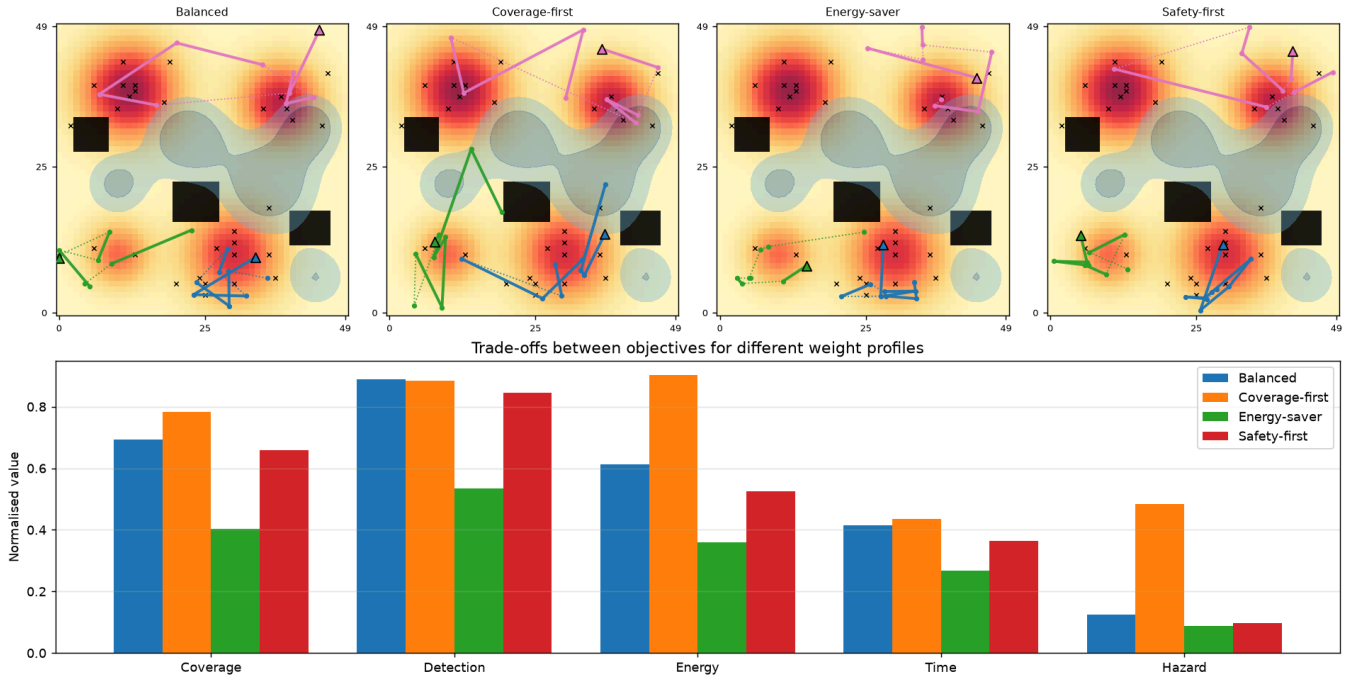


Figure 6: Trade-offs between objectives for four different weight profiles.

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8. GA vs Random Search

Table 4: GA vs random search (5 seeds each)

Metric (mean ± std)	GA (mut 0.12)	Random search
Fitness	0.362 ± 0.021	0.258 ± 0.011
Coverage	0.746 ± 0.027	0.677 ± 0.063
Detection	0.937 ± 0.051	0.770 ± 0.041
Energy (norm.)	0.534 ± 0.059	0.650 ± 0.074
Hazard (norm.)	0.145 ± 0.061	0.297 ± 0.160
Time (norm.)	0.318 ± 0.034	0.300 ± 0.032

The worst GA run beats the best random run. GA improves fitness by ~40 % (or ~68 % with mutation 0.05), with higher coverage and detection, less energy, and half the hazard exposure. The five GA runs reach similar fitness through different route topologies — coverage maps overlap at Jaccard 0.65 (vs 0.47 for random search).

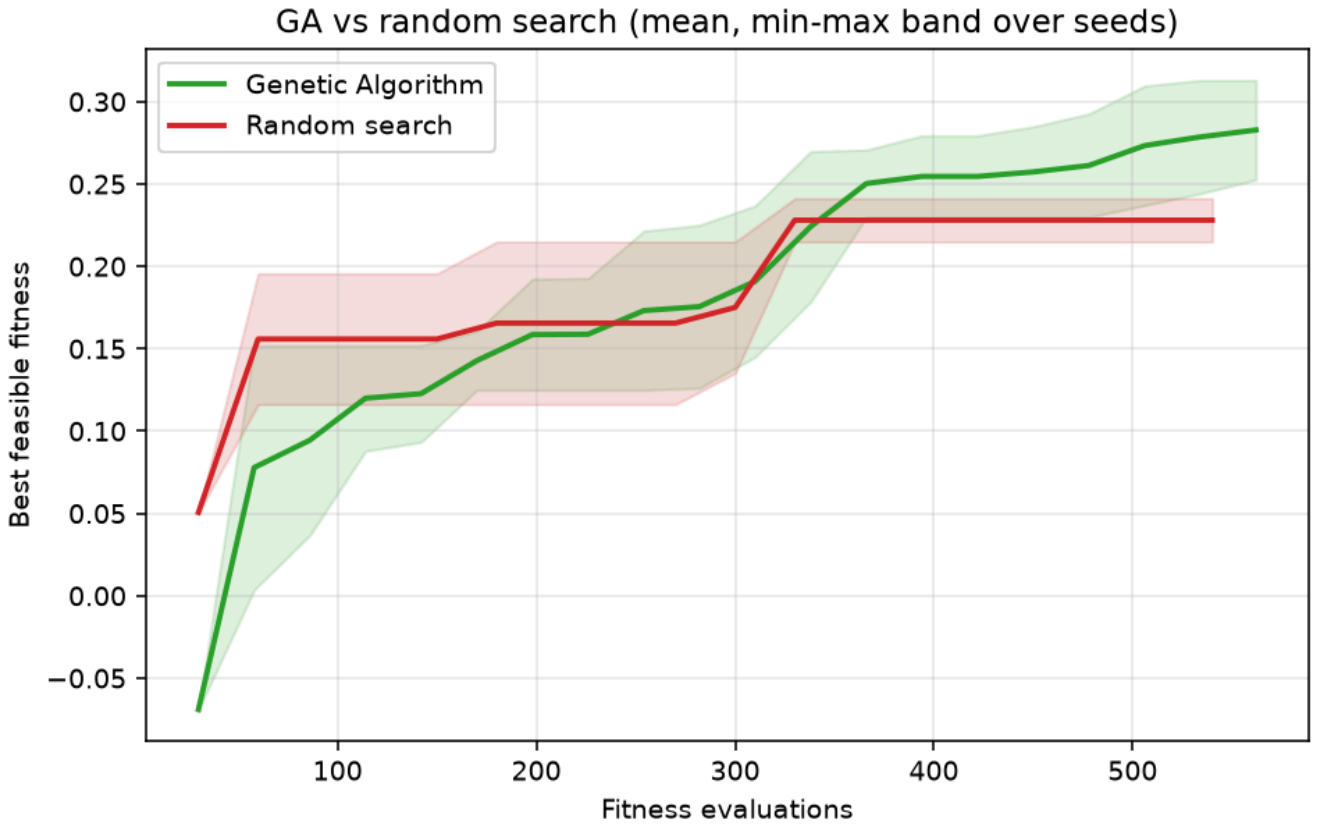


Figure 7: GA vs random search — best feasible fitness over evaluations.

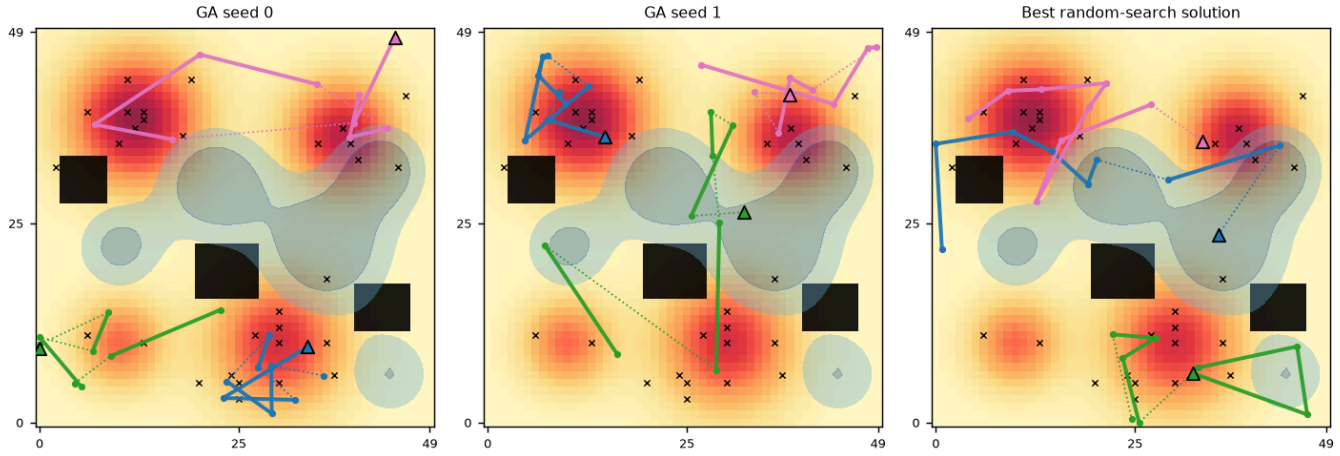


Figure 8: Five GA runs plus one random-search solution — different topologies, similar fitness.

9. Limitations and Future Work

Weighted-sum scalarisation gives one trade-off per run; NSGA-II would return the whole Pareto front. Segments are straight, hazards static, no collision avoidance. Extensions: NSGA-II, time-varying hazards, re-planning on drone failure.

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10. Reproducibility

`python main.py` regenerates every figure and `results/results.json` (all seeded).

GitHub Repository: https://github.com/janeetakhalil-collab/uav_ga_project